k = 4 index calculus on $E(\mathbb{F}_{p^4})$, measured

relations $\propto p^{1.00}$, Wiedemann $\propto p^{2.04}$, rho $\propto p^{1.93}$; crossover from the fits 2^{100} ($h = 4$), sensitivity 2^{86}–2^{125}

`research/gaudry_quartic_d4_exponent_20261015` · 2026-10-15 · class: measurement

Summary. Gaudry/Diem index calculus with FGHR's D_4 oracle (PR #1863) and rho with the negation map were run end to end in ecbench on `fp4_torsion` curves for 12 primes $p = 29 \ldots 1009$: 288 records, all verified ($[d]G = Q$), `S` a count. Fitted over the rungs, the relation phase grows like $p^{1.00}$ (95 %: 0.74–1.26), the linear algebra (Wiedemann, new here) like $p^{2.04}$ (1.85–2.22), and rho like $p^{1.93}$ (1.77–2.08), as the analysis says (q^1, q^2, q^2). Index calculus still loses to rho at every measured size ($9.6 \cdot 10^5\times$ at $p = 29$, $6.7 \cdot 10^4\times$ at $p = 1009$). From the measured fits, it meets rho at a group of $2^{100.4}$ ($h = 4$; bootstrap $2^{98.0}$–$2^{103.6}$); with the relations slope at its interval ends, $2^{85.9}$–$2^{124.9}$. The §11.16 model gave $2^{105.4}$–$2^{106.5}$. **This extrapolates 15 bits in p, 60 bits in group size, beyond the top rung.** At that point the relation phase is still 97 % of the cost. Double large primes cut the linear-algebra slope to 1.57 (theory 1.5) but cost 1.1–1.8× more on paired walks, and move the fitted crossover later (2^{124}).

1. Method

Protocol `PROTOCOL.md`, registered at `08722a2b7` before any session; amendments: (1) queue reordered and stop extended to 16 h (runtime, before any IC record); (2) audit scope by size; (3) relaunch of 20 jobs after the container's CPU set shrank. Code `0bea75038`. Arms (all `fghr-y-line:zero=1`, oracle `gaudry-d4:macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1`): `ic-d4-w` (`linalg=wiedemann`), `ic-d4-lp2`/`ic-d4-lp1` (`large_primes=2/1`, small base the first $\lceil N^\theta \rceil$ columns, $\theta = \frac{3}{4}, \frac{7}{9}$), `rho-neg`. One spec per (arm, p, target); the IC arms on a target share the walk. 59 of 72 jobs ran; 13 (third plain targets at five small rungs, second LP2 targets) did not start before the stop.

New code. `linalg=wiedemann` solves for the target through a left-kernel vector of the relation matrix, $O(m \cdot \mathrm{nnz} + m^2)$ multiplications mod r, priced as `row_ops`; the default dense elimination ran on the same rows as an unpriced shadow and **gave the same logarithm in all 48 IC runs**. The large-prime wrapper works on any summand oracle; the existing large-prime code is binary-field only.

Fits. OLS of $\ln(\text{rung mean})$ on $\ln p$; interval = union of the Student-t 95 % interval and a bootstrap (10,000, targets resampled within rungs). The crossover solves $\sum_k A_k p^{\beta_k} = \overline{S}_{\mathrm{rho}} \sqrt{p^4/h}$.

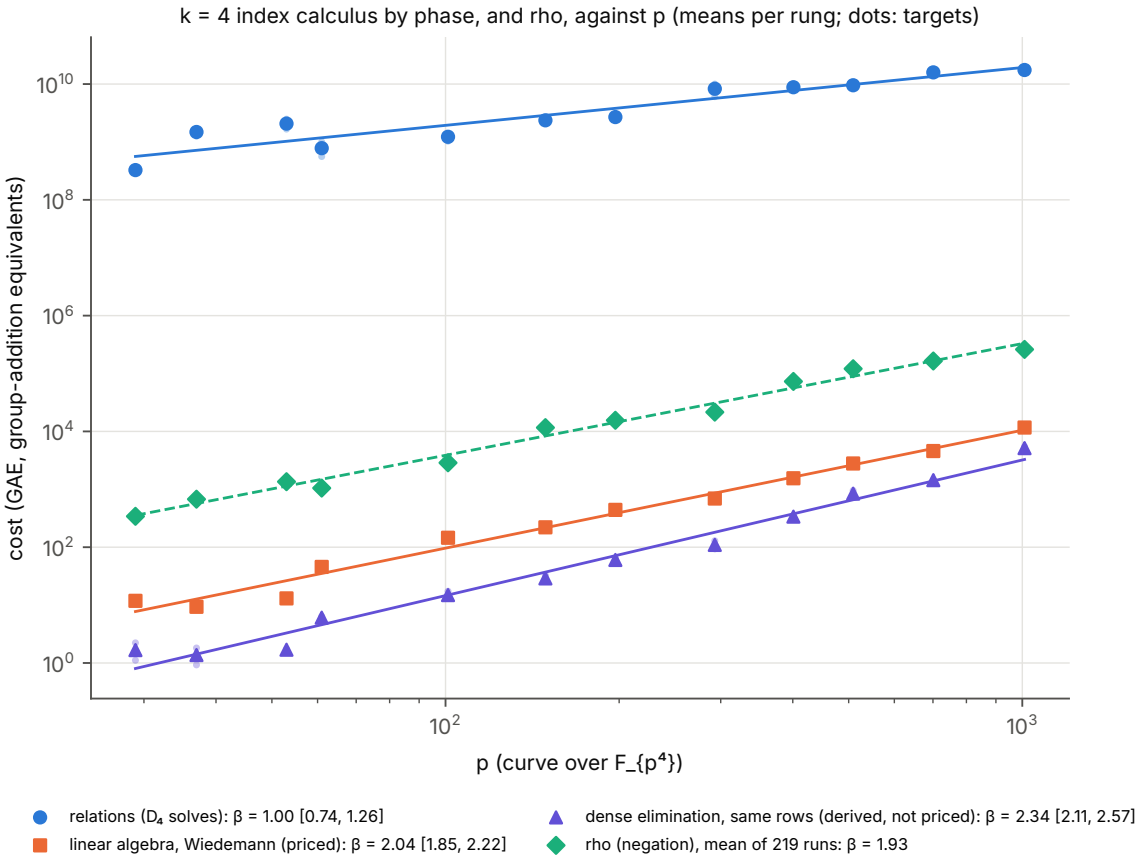


Figure 1: Cost by phase and rho against p (rung means; light dots: single targets). The dense elimination is derived from its `row_ops` at the same price and is not part of `S`.

2. Results

quantity	β	95 % interval	expected
relations phase (= total)	**1.00**	0.74–1.26 (bootstrap 0.96–1.04)	q^1
trials	0.90	0.64–1.15	
linear algebra, Wiedemann	**2.04**	1.85–2.22	q^2
dense elimination, same rows (derived)	2.34	2.11–2.57	q^3 worst case
rho (219 distinct runs)	**1.93**	1.77–2.08	q^2

The relations interval is wide because trials per relation vary between curves (60 at $p = 29$, 76 at 61, 310–340 at 37 and 53, 81 at 1009), which the within-rung bootstrap does not see. One D_4 solve costs $6.1 \cdot 10^5$ GAE at $p = 29$ and $8.3 \cdot 10^5$ at $p = 1009$; 106 of 202,677 systems (0.05 %) were degenerate and are charged. $\overline{S}_{\text{rho}} = 1.08$ (1.02–1.15), drifting from 1.6 at $p = 29$ to 0.9–1.0 at the top.

crossover (group size)	$h = 2$	$h = 4$	$h = 16$
measured fits, point	$2^{98.4}$	$\mathbf{2^{100.4}}$ ($p^* = 2^{25.1}$)	$2^{104.6}$
bootstrap within rungs	$2^{96.0}$–$2^{101.4}$	$2^{98.0}$–$2^{103.6}$	$2^{102.0}$–$2^{107.9}$
relations slope at 0.74 / 1.26	$2^{84.3}$–$2^{122.1}$	$\mathbf{2^{85.9}}$–$\mathbf{2^{124.9}}$	$2^{89.3}$–$2^{130.6}$
§11.16 model (#1863)	$2^{103.0}$–$2^{103.7}$	$2^{105.4}$–$2^{106.5}$	—

With rho's own fitted power law in p: $2^{108.7}$ ($2^{103.5}$–$2^{116.1}$). The top rung is a $2^{39.9}$ group, so the crossover is **60 bits of group size (15.1 bits of p) beyond the measurements**. The honest interval contains the model's figure. With the dense elimination priced instead, the cost meets rho at $2^{101.8}$ ($h = 4$) and rises above it again at 2^{126}; at $h = 16$ it never meets it: Wiedemann's q^2 is what keeps the window open.

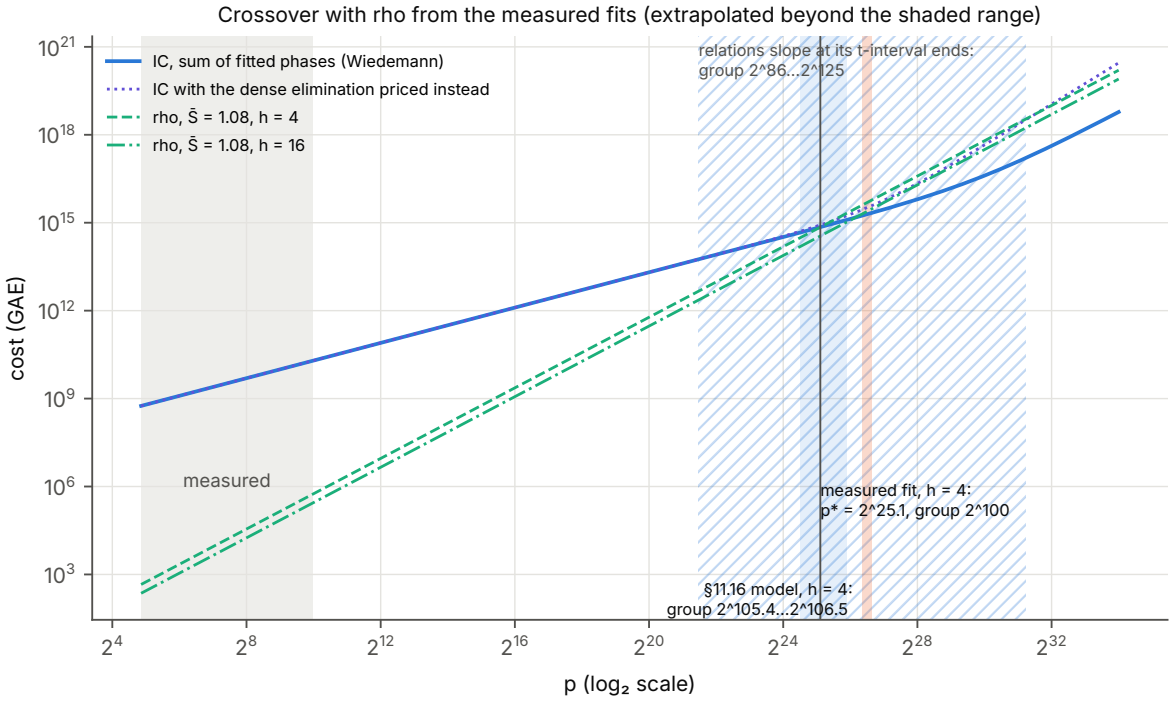


Figure 2: Extrapolation of the fitted phases to the crossover with rho. Grey: the measured range. Blue band: bootstrap; hatched: relations slope at its interval ends; orange: the §11.16 model at $h = 4$.

3. Large primes

arm	runs (rungs)	lin. alg. β	relations β	crossover $h = 4$
none	27 (12)	2.04 (1.85–2.22)	1.00 (0.74–1.26)	$2^{100.4}$
two, $\theta = \frac{3}{4}$	9 (8, $p \le 293$)	**1.57** (1.12–2.01)	1.23 (0.55–1.91)	$2^{123.9}$
one, $\theta = \frac{7}{9}$	12 (7, $p \le 149$)	1.62 (1.11–2.14)	1.49 (0.43–2.55)	2^{178} (153–214)

Paired S ratios (same target and walk), two large primes: 1.00, 1.00, 1.12, 1.36, 1.32/1.09, 1.22, 1.80, 1.44 for $p = 29$…293; one: 0.89–1.12 at $p \le 53$, 1.78–3.45 at $p \ge 61$. The matrix shrinks as the analysis says (Wiedemann 4.7× cheaper at $p = 293$) but it is

10^{-6} of the cost here and 3 % at the crossover; the extra trials land on the relations phase. Large primes would pay only where the linear algebra dominates, past about 2^{120}.

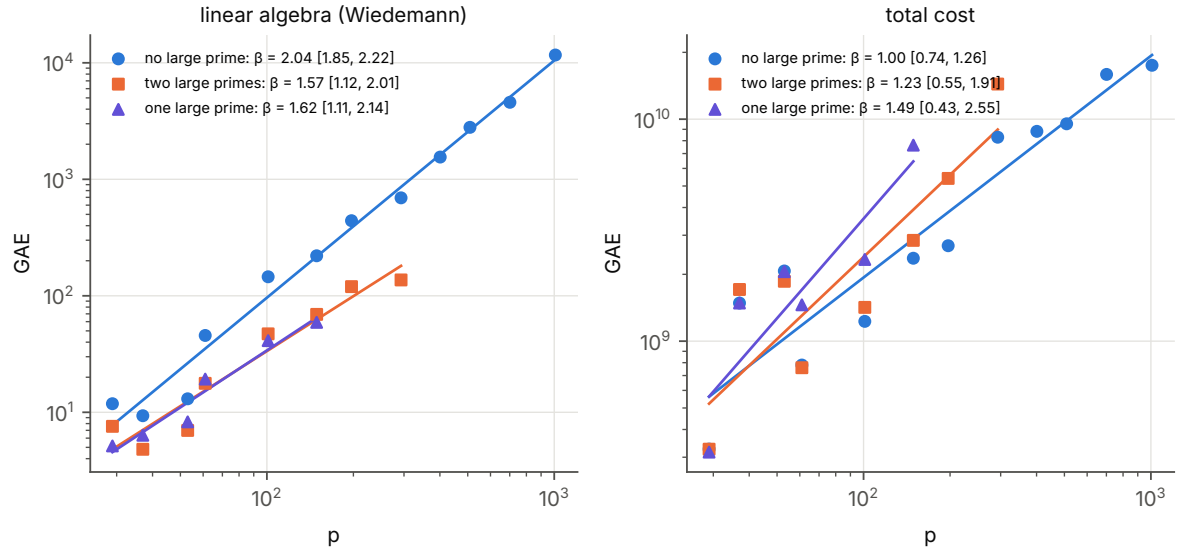


Figure 3: Linear algebra and total cost with and without large primes.

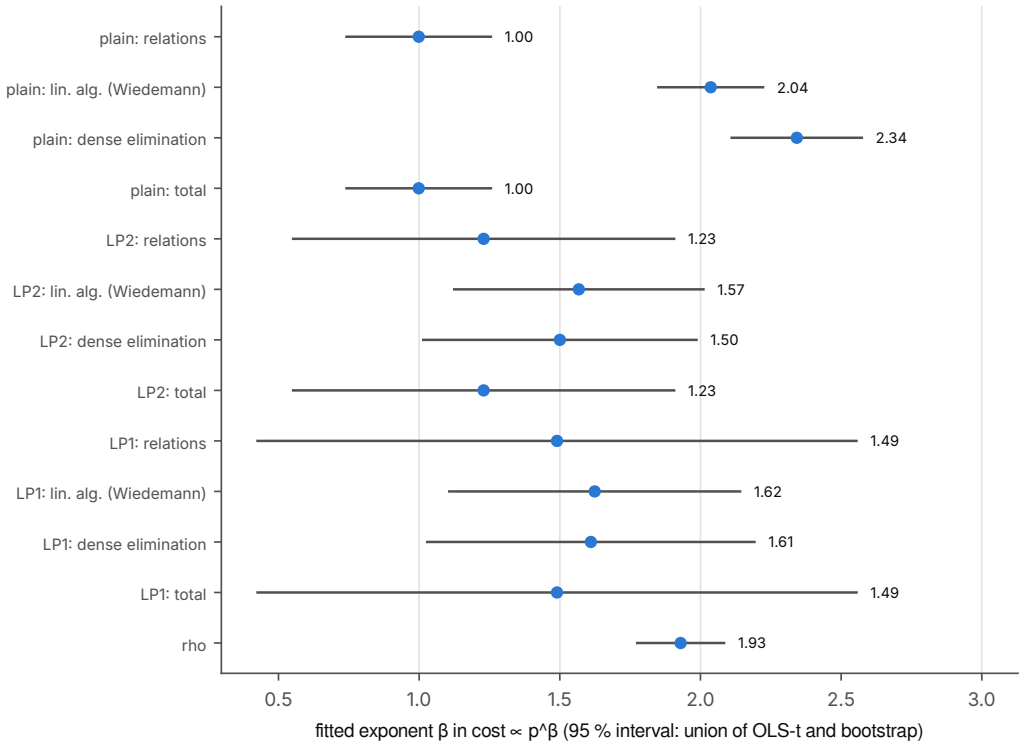


Figure 4: All fitted exponents with their 95 % intervals.

4. Predictions

P1 held: 288/288 verified, 268/268 replays reproduced. P2 held ($1.00 \in [0.9, 1.3]$). P3 held (2.04). **P4 not held** (dense elimination $2.34 < 2.4$). P5 held (1.93; S_{rho} slope −0.15). P6: crossover $2^{100.4} \in [2^{88}, 2^{115}]$ held; **"no dense crossover below 2^{200}" not held** ($2^{101.8}$, recrossing at 2^{126}). P7 held (1.57; 1.23 > 1.00; every paired ratio at $p \geq 101$ above 1). P8 held.

5. Audit

`ecbench verify` on all 60 sessions (receipts `audits/<job>.audit.json`; scope fixed by amendment 2): **audit OK on all 60, 288 records, 0 problems.** Full replay (`--replay-all`) on 50 sessions, every session with $p \leq 197$, every rho-only session and the spot checks `main-p293-t0` and `main-p401-t0`: **268 replays reproduced exactly**. Integrity only (stored $[d]G = Q$ checks, every planned run recorded, hashes) on the other 10: `main-p1009-t0/t1`, `main-p701-t0/t1`, `main-p509-t0/t1`, `main-p401-t1`, `main-p293-t1/t2`, `lp2-p293-t0`.

6. Limits

The exponents are fitted over $p = 29 \ldots 1009$ (groups 2^{19}–2^{40}); the crossover is an extrapolation of those fits, not a measurement, and its width is set by curve-to-curve scatter in the relation rate. The large-prime fits rest on one or two targets per rung over $p \leq 293$. S is a count on one host class; wall time is not admitted. No `docs/bounds` domain exists for $\mathbb{F}_{p^4}$ curves, so there is no challenge verdict. Study files: `README.md`, `raw/summary.json`, `sessions/`, run report `reports/run-20261015T0457Z/`.